Free variables after capture-avoiding substitution when the substituted variable occurs freely

The Open Logic Project contributors

openlogic:lambda-calculus/syntax:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-syntax/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the abstraction computation and the application-case setup (definedness split and membership dichotomy) are verbatim; the author marks the variable case exercise and the final union algebra of the application case as left, both elementary consequences of the displayed induction hypothesis.

Editorial note. The label thm:infv names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Inductive definition of free variables. Syntax unit, Definition of FV (section free variables, definition label fv and its clauses). Unfolds FV of abstraction and application inside the displayed calculation.
- Side-condition machinery of capture-avoiding substitution. Same unit, Definition of substitution (defn:substitution and clause \ollabel{defn:substitution-4}). Reads the definedness of the substitution through to the binder case at lines 1283-1286.

Theorem 0.1. *If* $x \in \operatorname{FV} M$*), then* $\operatorname{FV} M[N/x] = (\operatorname{FV} M \setminus \{x\}) \cup \operatorname{FV} N$*, provided the left hand is defined.*

Proof. By induction on the formation of M.

1. M is a variable: exercise.
2. M is of the form PQ: Since $(PQ)[N/y]$ is defined, it has to be $(P[N/x])(Q[N/x])$ with both substitution defined. Also, since $x \in \operatorname{FV} PQ$, either $x \in \operatorname{FV} P$ or $x \in \operatorname{FV} Q$ or both. The rest is left as an exercise.

3. M is of the form $\lambda[y][P]$. Since $\lambda[y][P][N/x]$ is defined, it has to be $\lambda[y][P[N/x]]$, with $P[N/x]$ defined, $x \neq y$ and $y \notin \mathrm{FV}\, N$; also, since $y \in \mathrm{FV}\, \lambda[x][P]$, we have $y \in \mathrm{FV}\, P$ too. Now:

$$
\begin{aligned}
&\mathrm{FV}\,(\lambda[y][P])[N/x] = \\
&\qquad = \mathrm{FV}\, \lambda[y][P[N/x]] \\
&\qquad = \mathrm{FV}\, P[N/x] \setminus \{y\} \\
&\qquad = ((\mathrm{FV}\, P \setminus \{y\}) \cup (\mathrm{FV}\, N \setminus \{x\}) && \text{by inductive hypothesis} \\
&\qquad = (\mathrm{FV}\, P \setminus \{x, y\}) \cup \mathrm{FV}\, N && x \notin \mathrm{FV}\, N \\
&\qquad = (\mathrm{FV}\, \lambda[y][P] \setminus \{x\}) \cup \mathrm{FV}\, N
\end{aligned}
$$

□